



UNIVERSITY OF THE PUNJAB

Roll No.



Fifth Semester 2018
Examination: B.S. 4 Years Programme

PAPER: Parametric and Nonparametric Tests (Theory)
Course Code: STAT-301

TIME ALLOWED: 30 mins.
MAX. MARKS: 10

Attempt this Paper on this Question Sheet only.

OBJECTIVE TYPE

- I. To perform a Run test for randomness, the data must be:
(a) Qualitative (b) Quantitative
(c) Divided into at least two categories (d) Divided into exactly two categories
- II. If the critical region is located on one side of the sampling distribution of test-statistic, the test is called:
(a) One tailed (b) Two tailed (c) both (a) & (b) (d) none of these
- III. Level of significance α lies between:
(a) -1 and +1 (b) 0 and 1 (c) 0 and n (d) $-\infty$ to $+\infty$
- IV. The Mann-Whitney U test is preferred to a t-test when
(a) Data are paired (b) Sample sizes are small
(c) The assumption of normality is not met (d) Samples are dependent
- V. What is maximum number of runs possible in a sequence of length 7 using two symbols?
(a) 6 (b) 4 (c) 3 (d) 7
- VI. A hypothesis that specifies all the values of parameter is called:
(a) Simple hypothesis (b) Composite hypothesis
(c) Statistical hypothesis (d) None of the above
- VII. In a Chi-square distribution no expected frequency should be:
(a) More than 5 (b) Less than 5 (c) Equal to 5 (d) Both (a) and (b)
- VIII. Which of the following is not true of parametric statistics?
(a) They are inferential tests
(b) They assume certain characteristics of population parameters
(c) They assume normality of the population
(d) They are distribution-free
- IX. The formula for Kruskal Wallis test is based upon
(a) Means (b) Deviations (c) Ranks (d) Categories
- X. The range of F variable is
(a) $0 \leq F^2 \leq +\infty$ (b) $-\infty \leq F^2 \leq 0$
(c) $-\infty \leq F^2 \leq +\infty$ (d) None of these



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Fifth Semester 2018

Examination: B.S. 4 Years Programme

Roll No.

PAPER: Parametric and Nonparametric Tests (Theory) TIME ALLOWED: 2 hrs. & 30 mins.
Course Code: STAT-301 MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE TYPE

Q1. Write short answers on the following questions

- What are non-parametric alternative of the ANOVA, when samples are dependent?
- What is the Power of the test.
- Explain briefly, with example, what do you understand by goodness of fit test.
- Outline the procedure of the sign test
- State the important application of χ^2 distribution.

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Q2. The government awarded grants to nine different experimental stations of the agricultural department to test the yield capabilities of two varieties of wheat. Five acres of each variety are planted each station and the yield, in kilograms per acre recorded as follows:

Station	1	2	3	4	5	6	7	8	9
Variety 1	900	700	850	950	1025	750	875	790	900
Variety 2	1050	750	790	900	1150	800	825	825	1000

Test the hypothesis at $\alpha = 0.05$, that the average yields of the two varieties of wheat are equal against the alternative hypothesis that they are unequal, assuming the distribution of yields to be approximately normal.

(10)

Q3. Two varieties of tomato were experimented with concerning their fruit-producing abilities, measured in pounds. The following data were obtained:

Location	1	2	3	4	5	6	7	8	9	10
Variety A	3.03	3.10	2.35	3.86	3.91	1.72	2.65	2.30	2.70	3.60
Variety B	2.28	3.68	2.17	3.56	3.73	1.85	1.48	1.86	2.76	2.68

Apply (i) the sign test, (ii) the Wilcoxon signed-rank test at the 0.05 level of significance, to test the hypothesis that there is no difference in fruit-producing abilities of the two varieties.

(10)

Q4. The various sums for two sets of data, each of 4 observations are as follows:

Set:	ΣX	ΣY	ΣXY	ΣX^2	ΣY^2
A	8	37	76	18	349
B	15	47	179	59	557

- Find the estimates of β_1 and β_2 , the regression coefficients of two linear regression lines.
- Construct the 95% confidence interval for $\beta_1 - \beta_2$ and test the hypothesis that $\beta_1 = \beta_2$ against $H_1: \beta_1 \neq \beta_2$ at the 0.05 level of significance.

(10)